Building Your Next .NET MAUI App

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David Ortinau

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25 yrs. web, interactive, mobile

Xamarin MVP 2013 - 2017

BA English, Maryville University

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.NET MAUI is the most productive way to develop native apps that perform great on any device that runs Android, iOS, macOS, or Windows from a single codebase.

Agenda

- Preamble
- Code
  - File > New
  - Navigation
  - Building UI
  - Customizing Platforms
- Recap
- Resources



Not Covered

- How this compares to Xamarin
- Upgrading from Xamarin

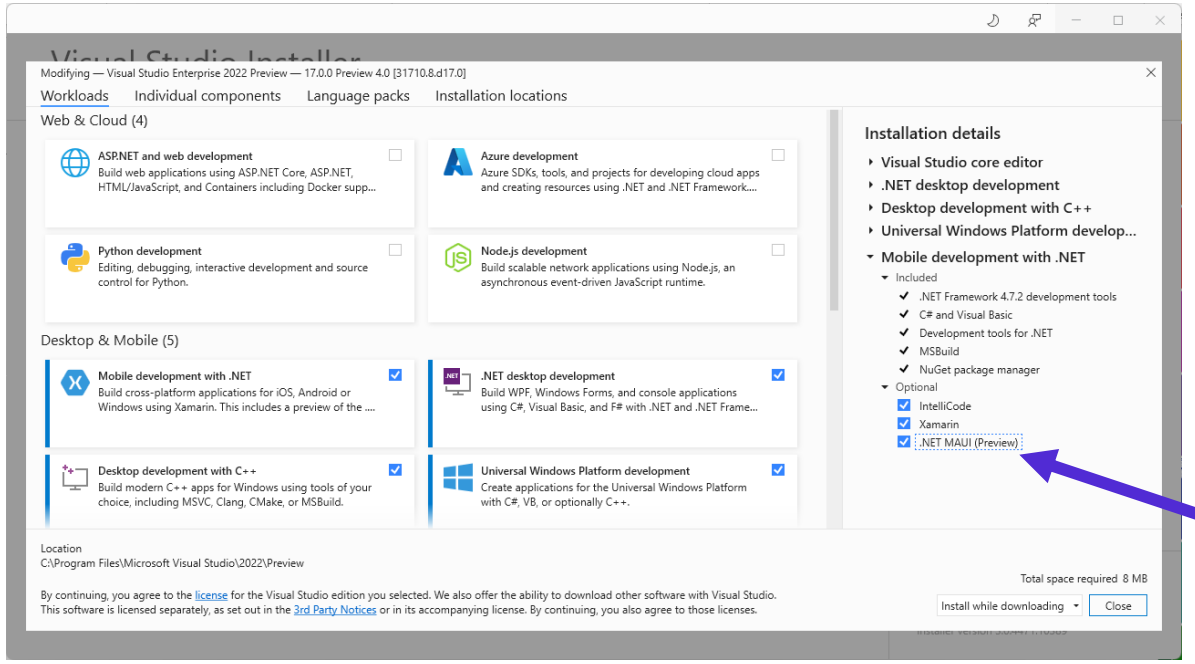


Preamble



.NET 6 & Visual Studio 2022

- Install Visual Studio 2022 and [redacted]NET MAUI (Preview)
- .NET MAUI is .NET
  - Ships with .NET
  - Is an optional workload
  - Has rudimentary CLI support with dotnet



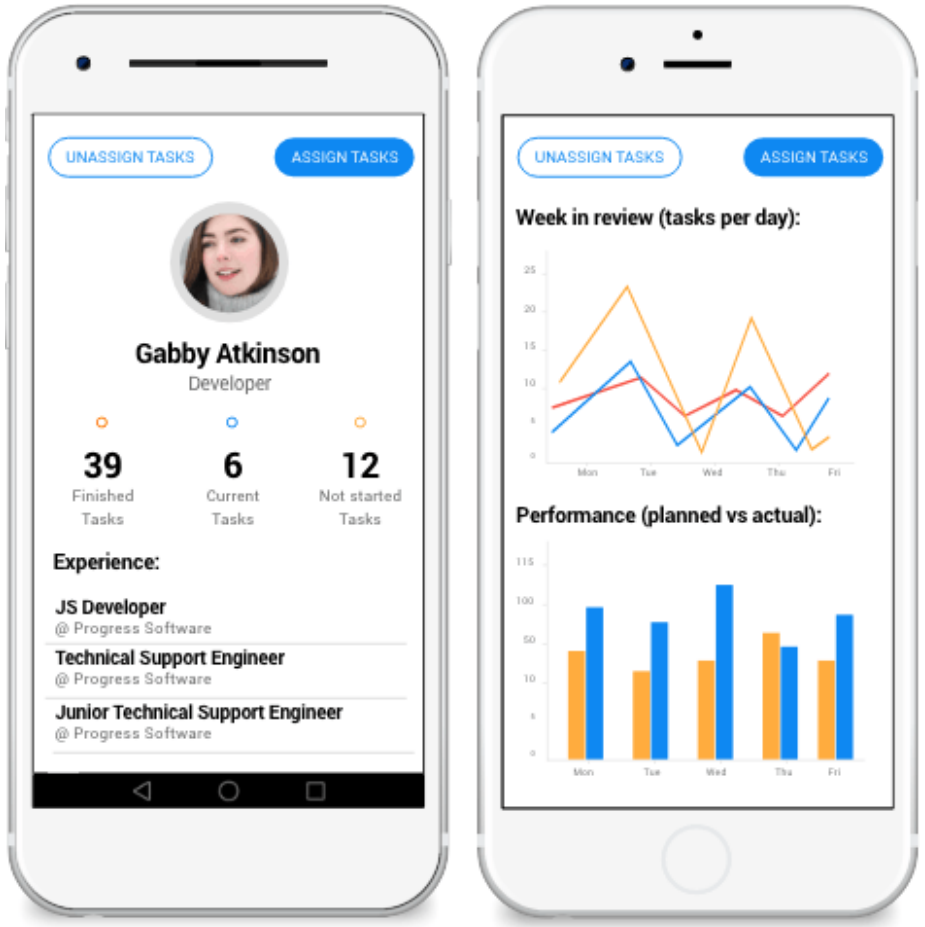


3rd Party Libraries

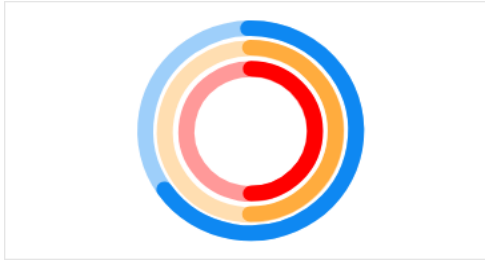
- Must be .NET 6 compatible
  - Or .NET Standard
  - Without Xamarin.Forms types
- Xamarin.CommunityToolkit -> .NET MAUI CommunityToolkit
- Xamarin.Essentials -> .NET MAUI (Microsoft.Maui.Essentials)

- Is what you need available?
- Can you do without?
- Can you write your own?

Progress Telerik

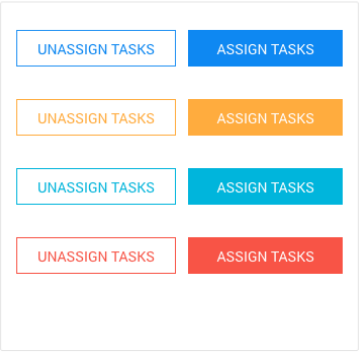






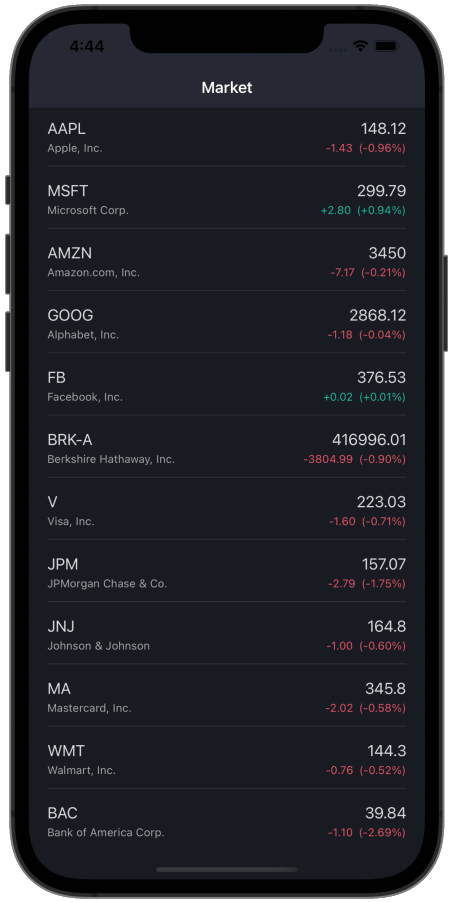
ID	Title	Assigned To
8207	UI Tests Builds timeout	Kian Parker
8211	Setup build machines	Davis Klein
8245	Add Barcode	Taylor Reed

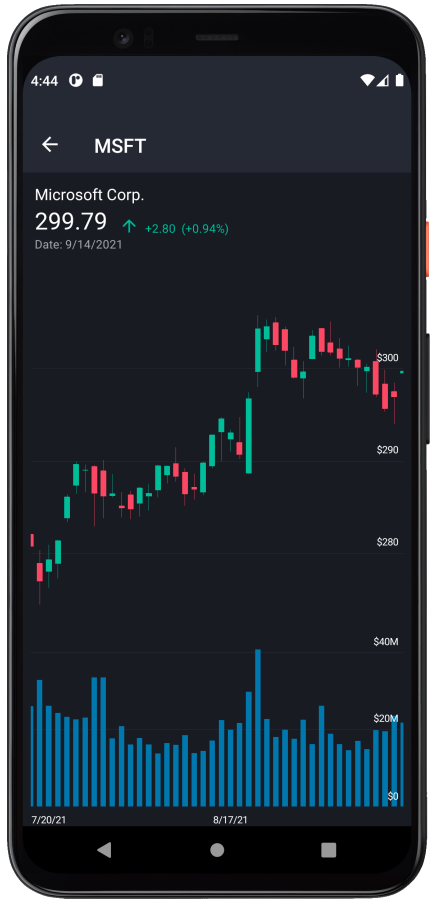




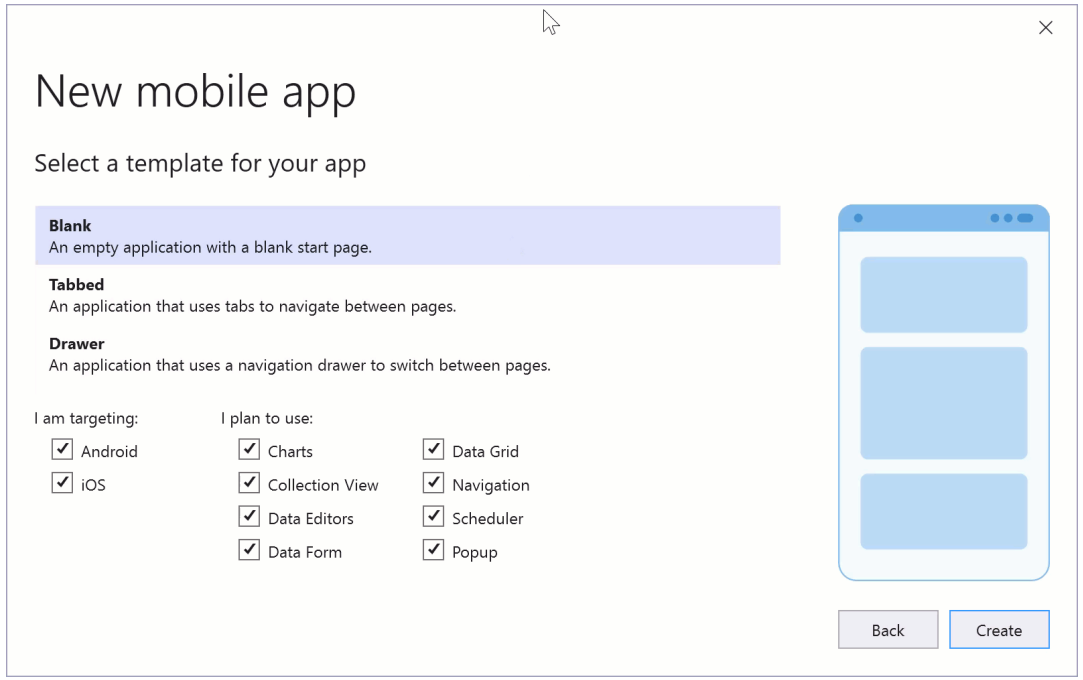
https://www.telerik.com/maui-ui

DevExpress Preview 8 Support









https://community.devexpress.com/blogs/mobile/archive/2021/09/30/net-maui-free-early-access-preview-of-data-editors-and-more-for-mobile-development-v21-2.aspx

- CollectionView
- ChartView
- PieChartView
- DrawerView
- DrawerPage
- TabView
- TabPage
- DataGridView
- TextEdit
- PasswordEdit
- MultilineEdit
- ComboBoxEdit
- AutoCompleteEdit
- NumericEdit
- DateEdit / TimeEdit
- CheckEdit
- SimpleButton

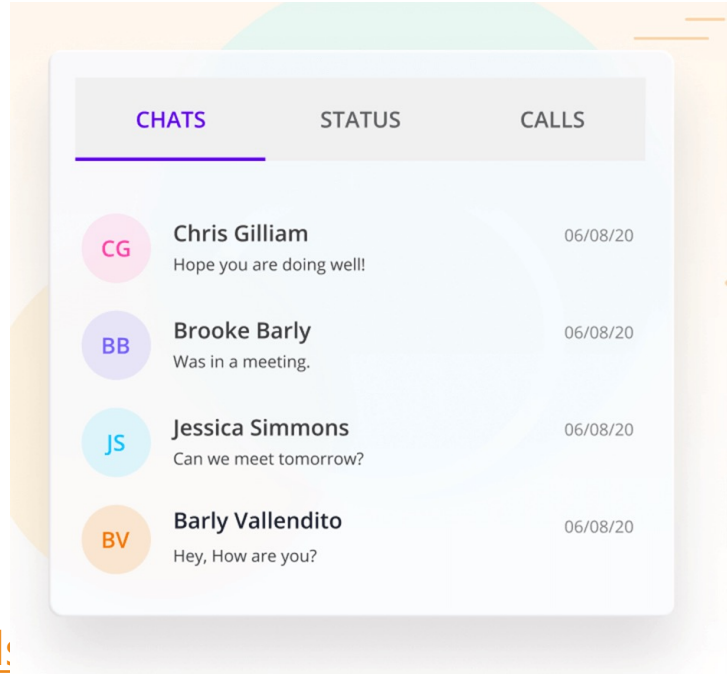
Syncfusion

[Introducing the First Set of Syncfusion .NET MAUI Controls | Syncfusion Blogs]



Are these controls migrated from Xamarin.Forms?

No. These controls were developed from scratch with the .NET MAUI graphics library and framework layouts, themselves with improved APIs and performance.





https://www.syncfusion.com/maui-control



.NET MAUI is Preview

- Quality
- Performance
- App size
- Controls and Layouts
- Features

- Are you good with it?



Pending UI Control Features for GA



Control	Feature
Button	BorderColor
	BorderWidth
	ContentLayout
	CornerRadius
	ImageSource
Editor	AutoSize
	IsTextPredictionEnabled (WinUI)
Entry	CursorPosition (WinUI)
	HorizontalAlignment (WinUI)
	IsTextPredictionEnabled (WinUI)
	IsPassword (WinUI)
	VerticalTextAlignment
	SelectionLength(WinUI)
Frame	BorderColor
	CornerRadius
	HasShadow

Control	Feature
Label	FormattedText (WinUI)
	TextType
	VerticalTextAlignment
Picker	SelectedItem (WinUI)
	TextColor (WinUI)
	VerticalTextAlignment
RadioButton	CheckedChanged
	GroupName
	IsChecked
SearchBar	CancelButtonColor
	FontAttributes (WinUI)
	FontSize (WinUI)
	MaxLength (WinUI)
	SearchCommand (Android)
	SearchCommandParameter (Android)
	VerticalTextAlignment
Slider	ThumbColor (WinUI)
	ThumbImageSource

Control	Feature
Switch	OnColor (WinUI)
	ThumbColor (WinUI)
WebView	CanGoBack
	CanGoForward
	Cookies
	Source
	Eval
	EvaluateJavaScriptAsync
	GoBack
	GoForward
	Reload
Other Controls	CarouselView
	CollectionView
	IndicatorView
	Map
	RefreshView
	SwipeView

https://github.com/dotnet/maui/wiki/Status

Let’s Looksee at Code





Easily customize controls

```
#if ANDROID
Handlers.EntryHandler
    .EntryMapper[nameof(IEntry.BackgroundColor)] = (h, v) =>
    {
        (h.NativeView as global::Android.Views.Entry).UnderlineVisible = false
    };
#endif
```

#ifdef if platform specific

Access the handler from anywhere

Dictionary of all props

Easily customize controls

```
Microsoft.Maui.Handlers.LabelHandler.LabelMapper.ModifyMapping(nameof(ILabel.Font), (handler, view, action)
=> {
    // Don't let any Label's font be set to anything over 20
    if (view.Font.Size < 20)
    {
        action?.Invoke(handler, view);
    }
});
```

Recap

- One project, multiple platforms
- Optimized layouts
- Easily add borders and shadows
- Easily style controls, compose controls, extend platforms
- XAML and/or C#
- Visual Studio 2022
  - XAML and C# Hot Reload
  - Live Visual Tree
  - Live Preview

Resources



Resources

- Docs: https://docs.microsoft.com/dotnet/maui/
- Presentations: https://github.com/dotnet-presentations/dotnet-maui
- Wiki: https://github.com/dotnet/maui/wiki
- WeatherTwentyOne: https://github.com/davidortinau/WeatherTwentyOne



ANNOUNCING

Xcode 13 Shipped

iOS 15

iPadOS 15

Apple removed macOS

Visual Studio 2019 v16.11.4

Visual Studio for Mac v8.10.10.8



Upcoming Dates

- November 2021
  - .NET 6 GA
  - .NET MAUI Preview
- December 2021 – Q1 2022
  - .NET MAUI Previews monthly
  - .NET MAUI RC
- Q2 2022
  - .NET MAUI GA



Thanks for joining!

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